

# Week 13 Revision

**ENN581 Robot Motion, Control and Planning**

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School of Electrical Engineering and Robotics

## ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

# Exam Prep Tips: Week 13

**Use the Guide to Succeeding in Engineering Exams**

- Access **the guide** for strategies to prepare and strategies to use during the exam.




**Remember the exam purpose**

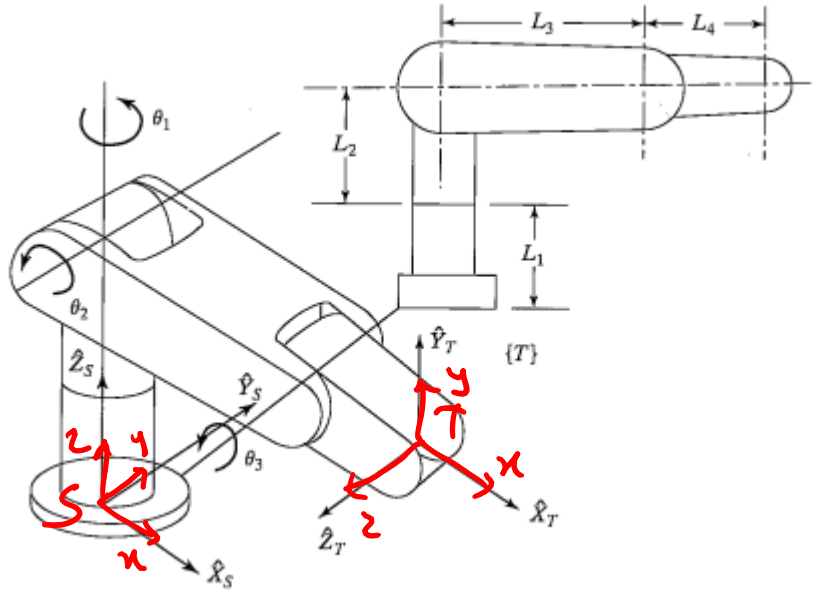
- It's a chance to demonstrate what you've learned in the unit. Consider your answers your chance to communicate what you know.
- Your lecturers and tutors have been setting you up for success. You will not get a question that you cannot solve using what has been taught in the unit.

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# Question 1: Forward kinematics

Write down the transformation matrix relating the tool frame {T} to the station frame {S} for the following RRR manipulator. Indicate intermediate frames.



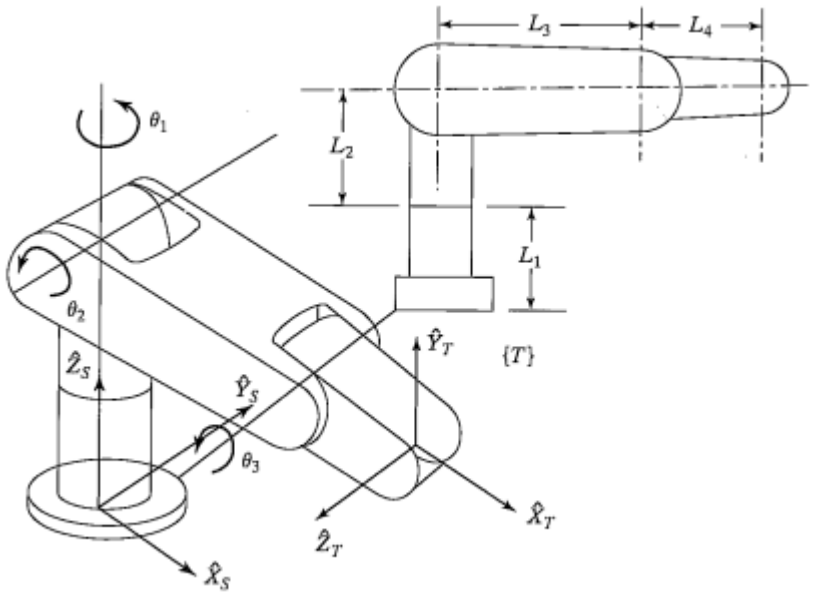

$$R_z(\theta_1)\, T_z(l_1+l_2)\, R_y(-\theta_2)$$

$$T_x(l_3)\, R_y(-\theta_3)$$

$$T_x(l_4)\, R_x(90^\circ)$$

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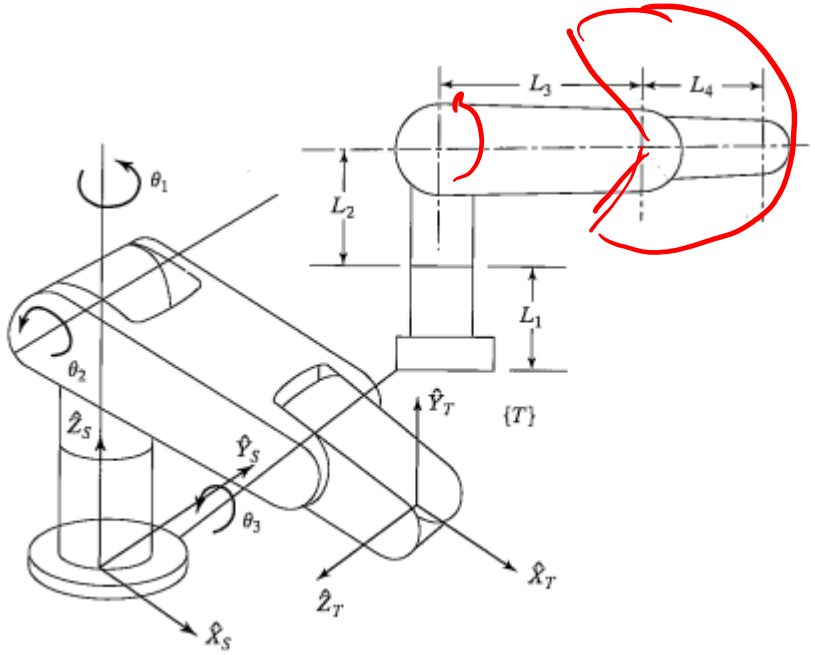
# Using elementary transform sequence

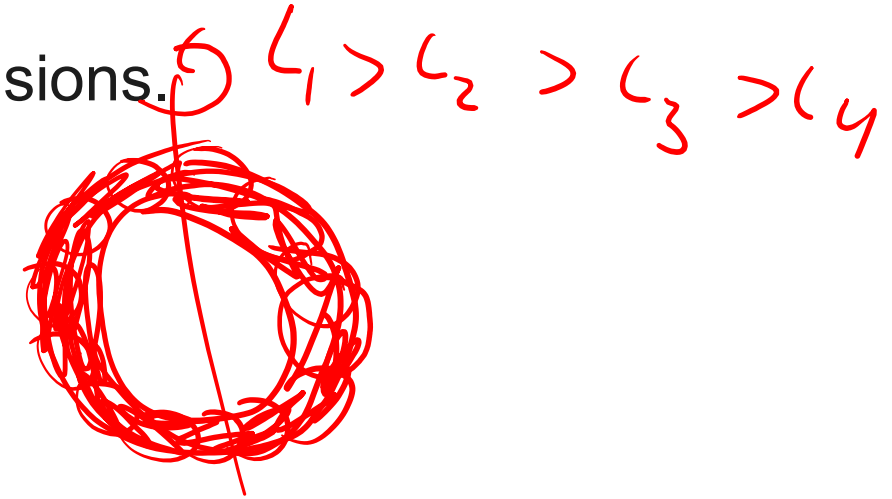



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# Question 2: Sketch the workspace of the manipulator

Assume no limit on joint angles, ignore collisions.



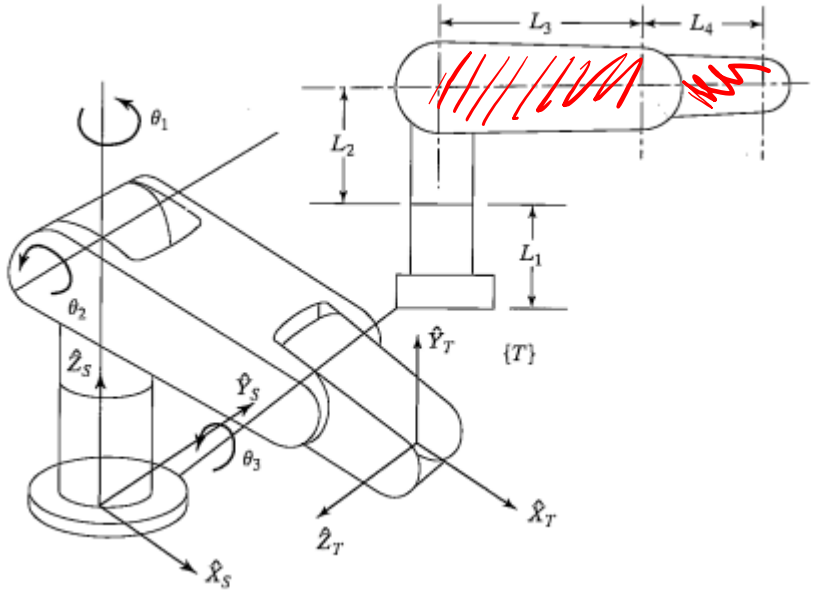

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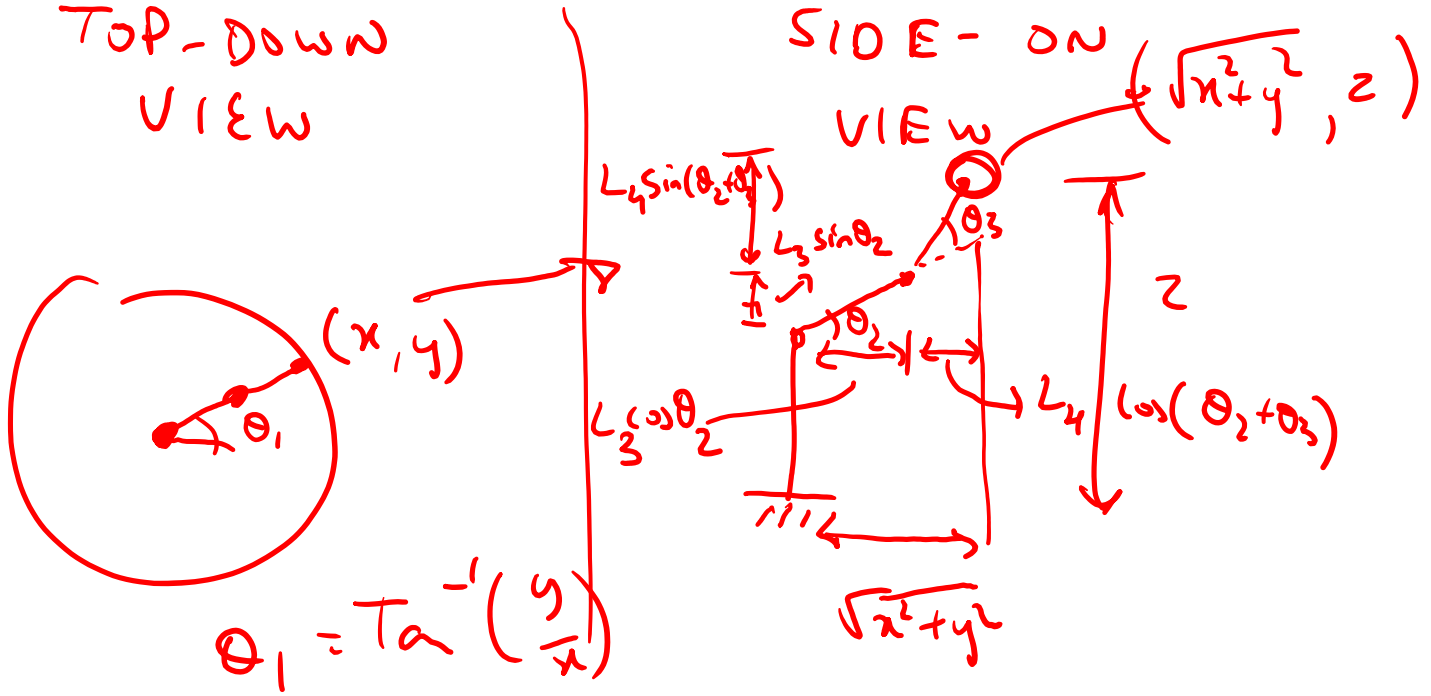
# Question 3: Derive the inverse kinematics

$$z = L_1 + L_2 + L_3 \sin\theta_2 + L_4 \sin(\theta_2 + \theta_3)$$

$$\sqrt{x^2+y^2} = L_3 \cos\theta_2 + L_4 \cos(\theta_2 + \theta_3)$$

You are given the position of the end effector $(x_T, y_T, z_T)$

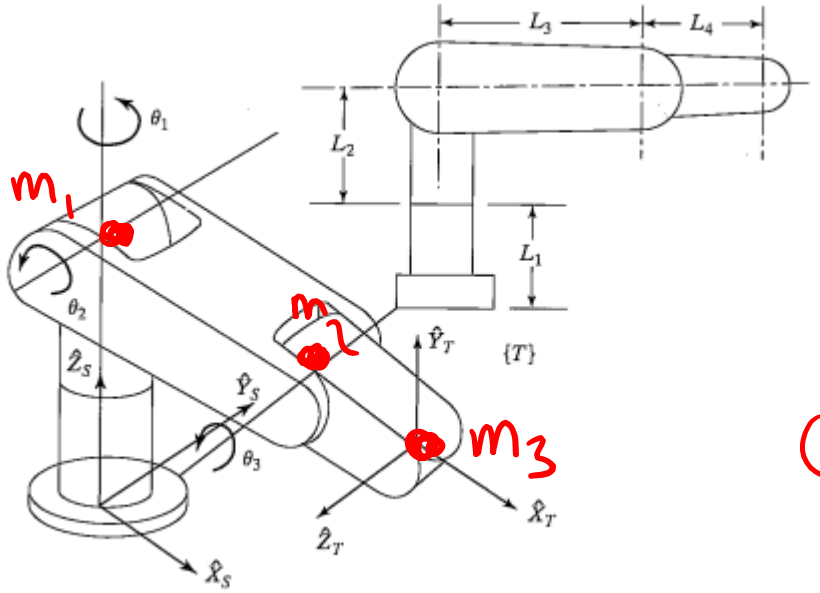



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# Question 4: Derive the equations of motion

mass, moments of inertia and positions of centre of mass are given.




$$\mathcal{L} = KE_1 - PE_1 + KE_2 - PE_2 + KE_3 - PE_3$$

$$Q = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q}$$

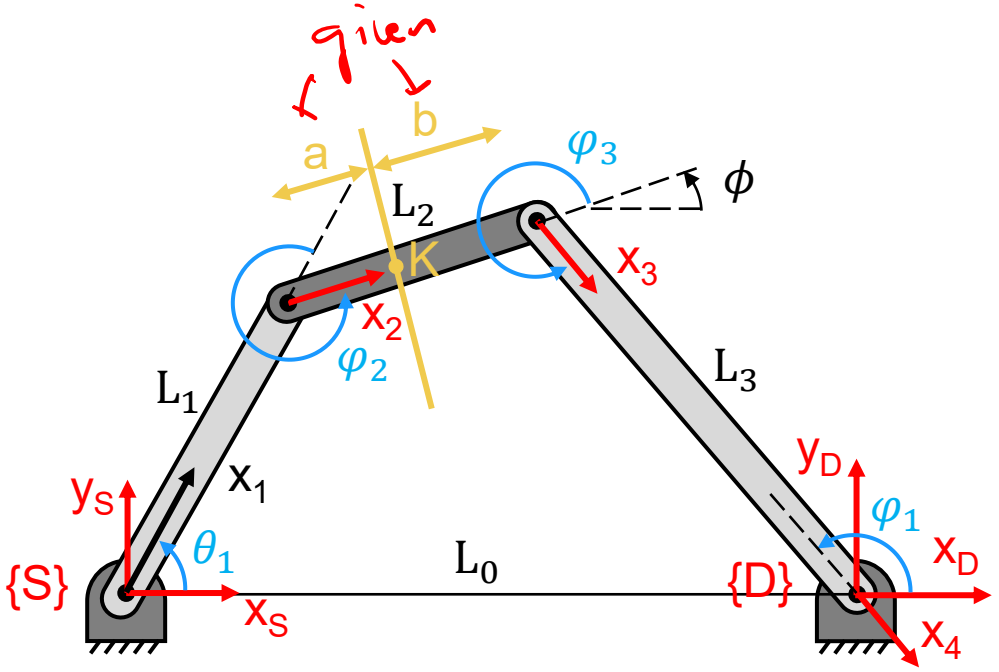
FK frames → derivatives → KE

→ PE

$$Q_1 = \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_1}\right) - \frac{\partial \mathcal{L}}{\partial q_1} \quad \Big| \quad Q_2 = \quad \Big| \quad Q_3 =$$

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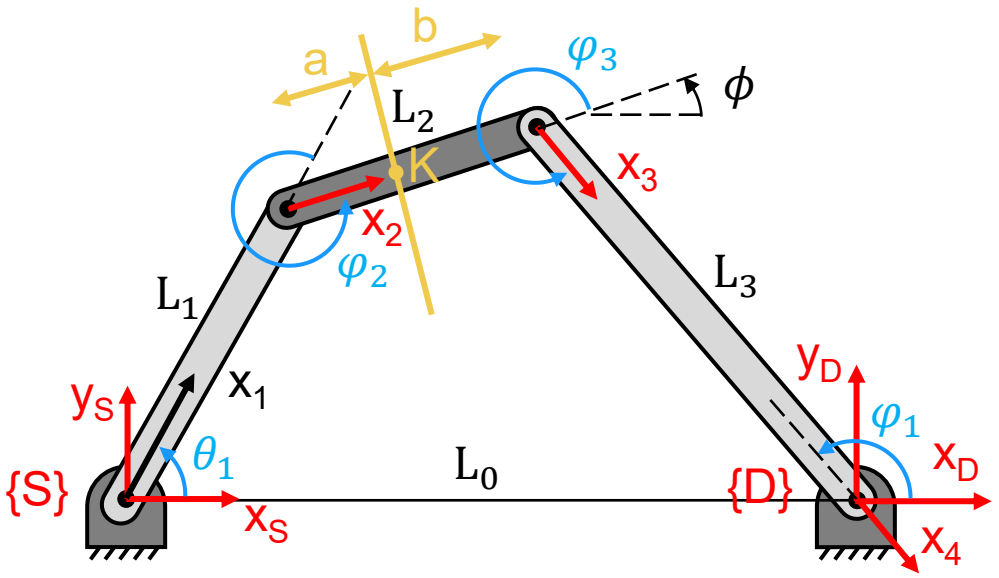
# Question 5: Parallel robots IK




- Given:
  - The position of a point K on the output link
  - The orientation $\phi$ of the output link (alternatively, ${}^{S}R_2$ of link 2 is given)
- Known:
  - $L_0$, $L_1$, $L_2$, $L_3$
  - $L_2 = a + b$
  - a and b are known
- Determine:
  - $\theta_1$, $\varphi_1$, $\varphi_2$ and $\varphi_3$

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# Start from what is given




$$x = L_1 \cos\theta_1 + a\cos(\theta_1 + \varphi_2)$$

$$y = L_1 \sin\theta_1 + a\sin(\theta_1 + \varphi_2)$$

$$\phi - \theta_1 = \varphi_2 - 2\pi$$

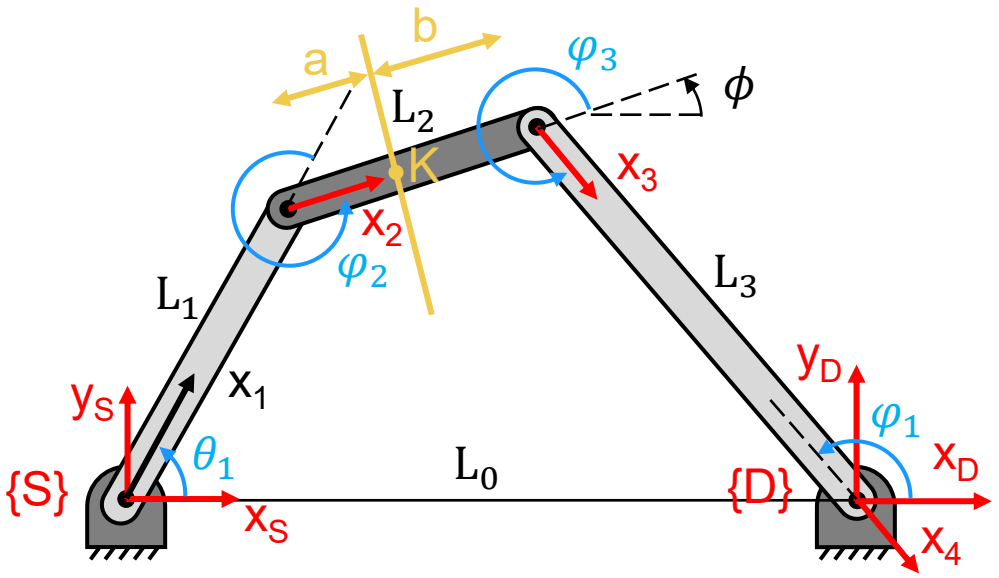
Solve for $\theta_1$ and $\varphi_2$

$$\theta_1 = \text{atan2}(y - a\sin\phi\,, x - a\cos\phi)$$

$$\varphi_2 = \phi - \theta_1 + \cancel{2\pi}$$

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# Four bar IK - continued




$$x = L_0 + L_3 \cos \varphi_1 + b \cos(\varphi_1 + \varphi_3)$$

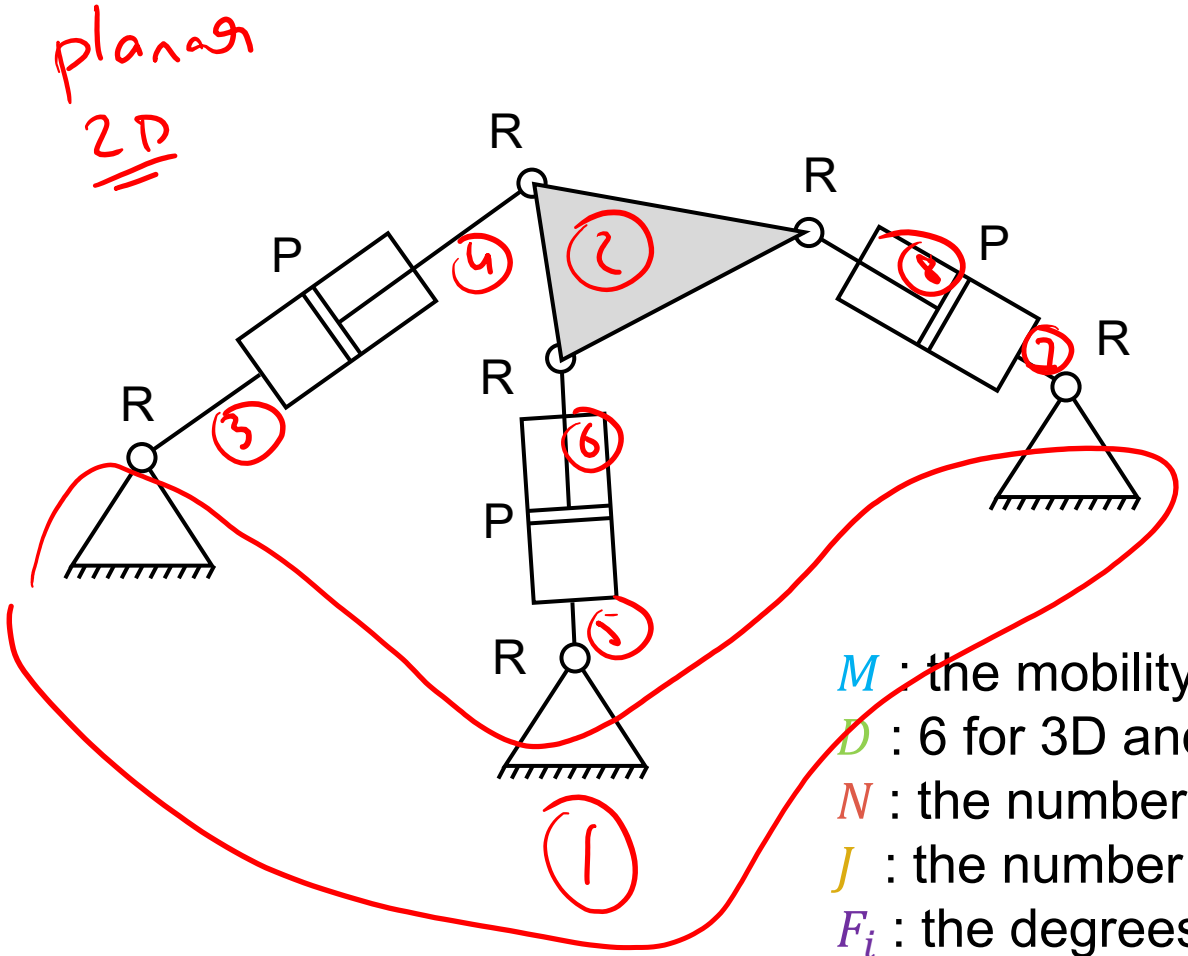
$$y = L_3 \sin \varphi_1 + b \sin(\varphi_1 + \varphi_3)$$

$$\varphi_1 - \phi = \varphi_3 - \pi$$

Solve for $\varphi_1$ and $\varphi_3$

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# Question 6: Mobility of parallel manipulators

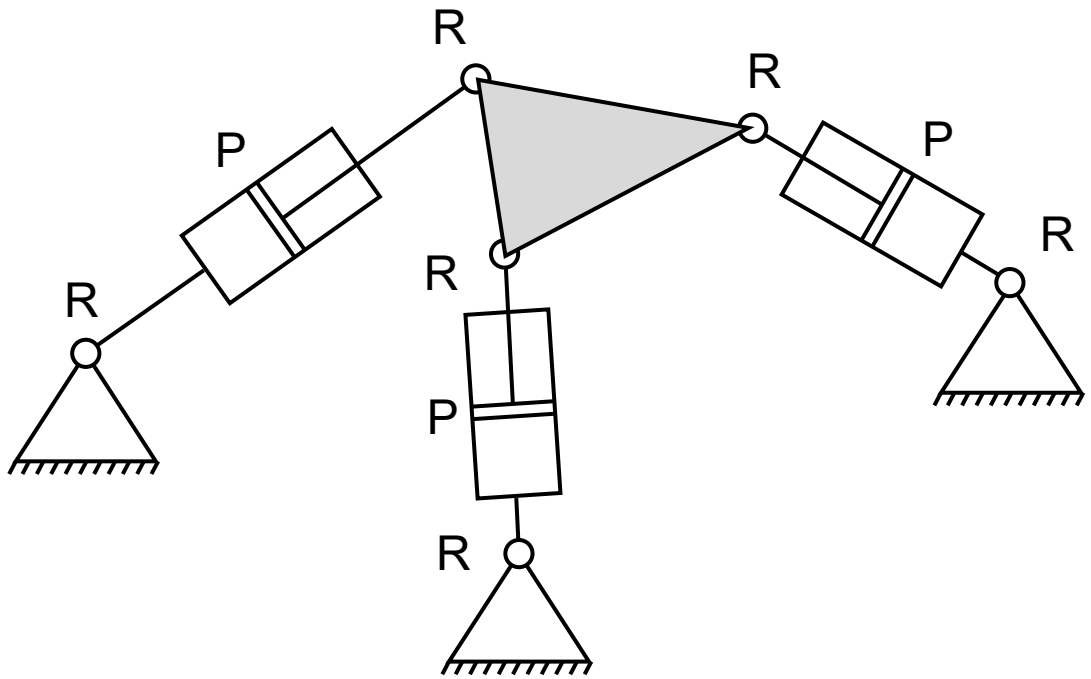



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$

$M$ : the mobility / DoF of the mechanism,
$D$ : 6 for 3D and 3 for 2D
$N$ : the number of links in the mechanism (one of which is grounded)
$J$ : the number of joints
$F_i$ : the degrees of freedom of joint $i$

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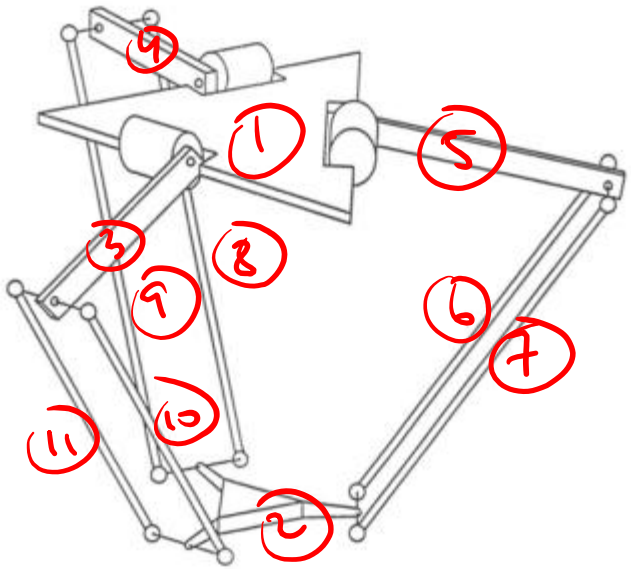
# Question 6: Mobility of parallel manipulators

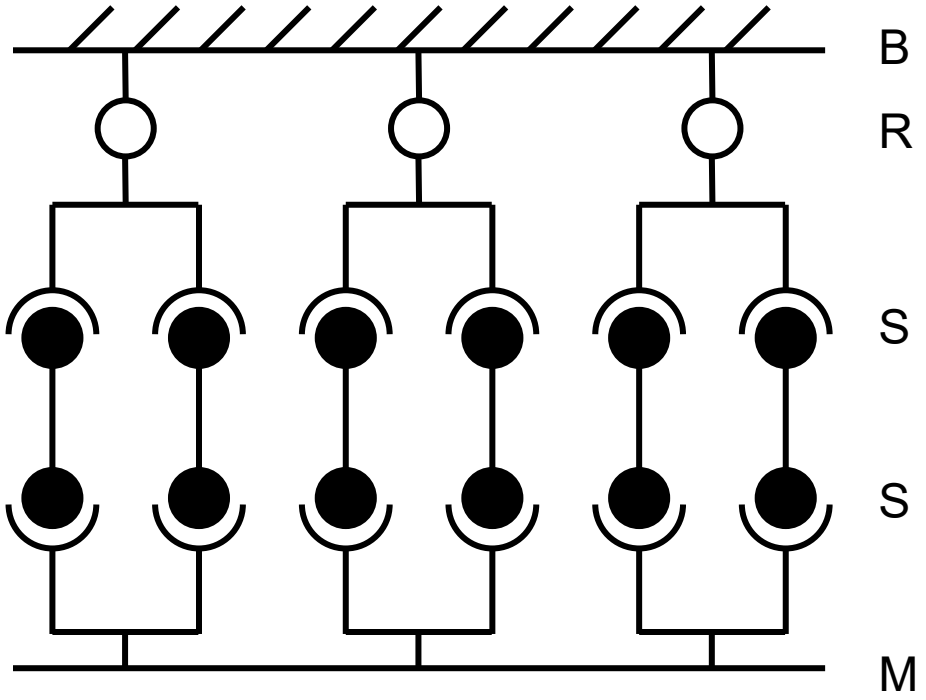



$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 3(8 - 9 - 1) + (9)$$
$$= 3$$

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# Question 7: Mobility of delta mechanisms

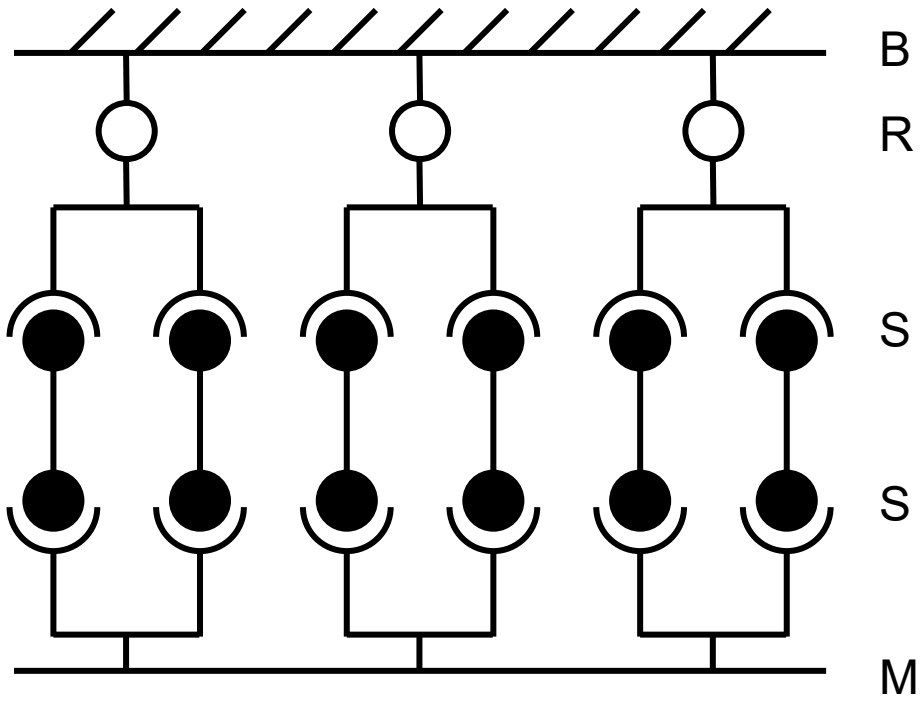



Note: the mechanism is still 3D despite this depiction

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# Apply the mobility equation to this mechanism




$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i$$
$$= 6(11 - 15 - 1) + (1+4\times3)\times3$$
$$= 9$$

Note: the mechanism is still 3D despite this depiction

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# Extended Grübler-Kutzbach equation

$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

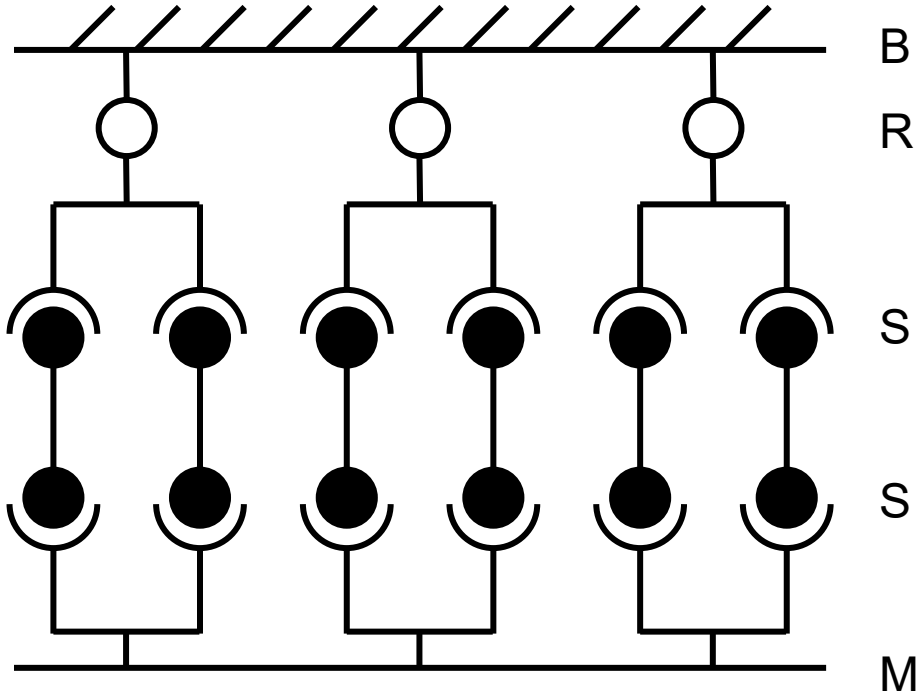
$M$ : the mobility / DoF of the mechanism,
$D$ : 6 for 3D and 3 for 2D
$N$ : the number of links in the mechanism (one of which is grounded)
$J$ : the number of joints
$F_i$ : the degrees of freedom of joint $i$
$R_C$ : number of over constraints
$R_M$ : local mobility / DoF

$$R_C = \sum_{i=1}^{J} F_i - \sum_{k=1}^{L} u_k$$

$u_k$ are unique DOFs from each closed loop $k$

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# Apply the extended mobility equation




$$M = D(N - J - 1) + \sum_{i=1}^{J} F_i + R_C - R_M$$

$$= 6(11 - 15 - 1) + (1+4\times3)\times3 + 0 - 6$$

$$= 3$$

R. S 3 axes

4 spherical

Note: the mechanism is still 3D despite this depiction

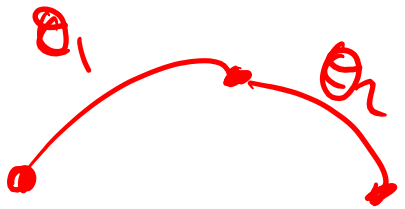
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# Question 8: Cubic paths

Solve for the coefficients of two cubics that are connected in a two-segment spline with continuous acceleration at the intermediate via point. The initial angle is $\theta_0$ the via point $\theta_v$ is and the goal point is $\theta_g$.

$$\theta_1(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

$$\theta_2(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3$$




$$\theta_1(t_0) = \theta_0$$

$$\theta_1(t_v) = \theta_v = \theta_2(0)$$

$$\ddot{\theta}_1(t_v) = \ddot{\theta}_2(0)$$

$$\theta_2(t_F) = \theta_g$$

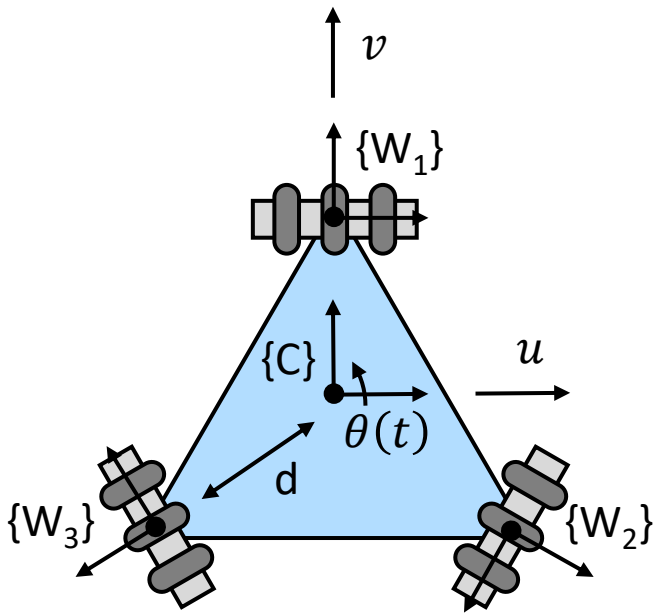
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# Question 9: Cubic paths

A prismatic joint, initially at rest, is programmed to move from an initial position d=0.02m to a resting goal position for d=0.15m in 0.8 seconds. Calculate the coefficients of a cubic spline that accomplishes this motion.

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# Question 10: Mobile robots




Consider the following robot with three omnidirectional wheels. Each of these can be driven independently. Their angular velocities are given as $\dot{\varphi}_1$, $\dot{\varphi}_2$ and $\dot{\varphi}_3$. The rollers are oriented perpendicular to the driving direction of each wheel.

The geometry of the robot is an equilateral triangle. The distance from the centre to each verted is given as d.

Derive a kinematic expression that relates the angular velocities of the wheels ($\dot{\varphi}_1$, $\dot{\varphi}_2$ and $\dot{\varphi}_3$) to the velocity of the robot chassis frame ($u, v, \dot{\theta}$).

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# The generalised model of a wheeled robot

$$\dot{\varphi}_i = \begin{bmatrix} \frac{1}{r_i} & \frac{\tan\gamma_i}{r_i} \end{bmatrix} \begin{bmatrix} \cos\beta_i & \sin\beta_i \\ -\sin\beta_i & \cos\beta_i \end{bmatrix} \begin{bmatrix} 1 & 0 & -{}^{C}y_{W_i} \\ 0 & 1 & {}^{C}x_{W_i} \end{bmatrix} \begin{bmatrix} u \\ v \\ \dot{\theta} \end{bmatrix}$$

$\dot{\varphi}_i$: the angular velocity of the wheel
$r_i$ : the radius of the wheel
$\gamma_i$ : the angle between $v_{free-roll}$ and the normal to the plane of the wheel
$\beta_i$ : the angle between the $v_{drive}$ and the robot chassis
${}^{C}x_{W_i}$ and ${}^{C}y_{W_i}$ : the position of the wheel with respect to the chassis
$\theta$ : the orientation of the robot chassis with respect to a global inertial frame of reference
$\dot{x}$, $\dot{y}$, $\dot{\theta}$ : the velocity of the robot chassis in the global inertial frame of reference

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# Any other questions?

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